

Hamish Chhagan

Full Stack Developer

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Projects

WriteLab — AI Content Generation & SEO Platform | 2025–2026

- Built a full-stack SaaS platform for AI-powered content generation, SEO optimization, and multi-channel publishing with React 19, Node.js/Express, PostgreSQL, and Redis.
- Implemented multi-LLM orchestration using LangGraph with 4 providers: Anthropic Claude, Google Gemini, DeepSeek, and Perplexity for research-augmented generation.
- Designed unified research interface reducing user workflow from 6+ page transitions to a single view with parallel API orchestration and progressive loading (SERP ~45s, briefs ~60s, topics ~120s).
- Built RAG system using pgvector for semantic search across website content and knowledge bases with OpenAI ada-002 embeddings.
- Integrated Google enterprise ecosystem via single OAuth flow: Search Console (keyword performance), Analytics 4 (traffic), and Business Profile (multi-location management).
- Developed plugin-free WordPress publishing using Application Passwords with draft/live publishing, AI-suggested categorization, and Yoast/Rank Math compatible SEO metadata.
- Implemented security: session-based auth (Passport.js), CSRF protection, SSRF prevention, rate limiting tiers, and encrypted credential storage.

E-Commerce Data Pipeline & Product Catalog System | Oct 2025 – Feb 2026

Carrel Electrade — NZ electrical components distributor

- Architected end-to-end ETL pipeline transforming raw product data from 3 sources into a production-ready WooCommerce catalog of 10,888 rows across 169 columns.
- Built 4-stage pipeline: PDF parsing (4,829 products from 134-page pricebook using pdfplumber + custom regex), web scraping (180 pages, 315 images, 156 spec PDFs), data enrichment, and WooCommerce conversion.
- Achieved 98.8% image coverage from only 315 unique images through intelligent series-level propagation (section → subsection → brand fallback).
- Engineered 400+ regex patterns for description splitting, 315+ categorization rules, and 27 SKU grouping patterns for automatic variable product detection.
- Integrated GPT-4o-mini for AI-assisted categorization with batch processing (20 products/call), validation against 155 categories, and rule-based fallback.
- Implemented resilient scraping: 1.5s rate limiting, persistent JSON cache (2.2MB), incremental saves, and 5-strategy SKU-to-image matching.

Marketing Mailer — SaaS Email Marketing Platform | 2025

- Built production-ready email marketing platform with Next.js 16, React 19, TypeScript, PostgreSQL (Prisma ORM), and Redis for queue management.
- Implemented multi-tenant architecture with user-provided API keys for complete tenant isolation, AES-256 encrypted credential storage, and per-user domain management.
- Developed queue-based email processing with BullMQ: rate limiting (10 emails/sec), 5 concurrent workers, retry logic, and job monitoring.
- Built real-time analytics pipeline with TinyBird integration tracking sent, delivered, opened, clicked, bounced, and complained events via webhook ingestion.
- Features: DNS-verified domain management (SPF, DKIM, DMARC), dynamic template builder with variable interpolation, CSV recipient import, and campaign workflow management.

AI-Powered Lead Generation Chatbot Platform

- Built multi-tenant chatbot platform with user authentication, bot management, and real-time chat interfaces using React, TypeScript, Node.js, Express, and PostgreSQL.
- Implemented RAG system using LangChain, FAISS vector store, and OpenAI embeddings for context-aware responses from uploaded PDF knowledge bases.
- Created dynamic tool generation system allowing chatbots to capture structured lead data based on customizable schemas, with automatic email notifications via Nodemailer.

- Migrated backend from Flask to Express.js, maintaining feature parity while improving performance and type safety with TypeScript.

The Phone Mason — Phone Repair Business Website

- Built complete business website from scratch including service pages, online booking system, and contact forms optimized for local lead generation.
- Implemented SEO strategy improving average search position from 28 to 15, achieving page 1 rankings for competitive keywords including 'phone repairs' (#10), 'samsung battery replacement' (#10), and 'android repair' (#9).
- Generated 920+ organic clicks and 150K+ impressions over 16 months through technical SEO, content optimization, and local citation building.
- Managed ongoing Google Ads campaigns, site performance, security updates, and content freshness for sustained lead generation.

Kiwi Kandy — WooCommerce E-commerce Platform

- Built complete WooCommerce e-commerce store from scratch for NZ confectionery retailer, including custom product configurations, shopping cart, and secure payment gateway integrations.
- Implemented SEO strategy improving average search position from 51 to 12 (39 positions), increasing impressions by 115% and clicks by 445% over 6 months.
- Integrated Google Merchant Center for automated product feed management, enabling product listings and inventory sync across Google Shopping.
- Developed custom shipping and fulfillment automation including label generation, inventory tracking, and customer notification workflows.

Work Experience

Fullstack Developer | August 2024 – Present

Rank / Web Merchants | Auckland CBD

- Built end-to-end ETL pipeline for Carrel Electrade, transforming 10,400+ products from 3 data sources (PDF pricebook, web scraping, inventory) into production-ready WooCommerce catalog with 98.8% image coverage.
- Developed custom LMS system for Test and Tag New Zealand, implementing compliance tracking, certification management, and automated reporting workflows.
- Led full website redesign for SmashedIt.co.nz, implementing modern UI/UX, payment system integration, and performance optimizations.
- Created new features for The Cylinder Guy e-commerce platform, including custom product configurators and checkout flow improvements.
- Implemented Google Tag Manager and GA4 reporting across client sites, enabling data-driven decision making with custom event tracking and conversion funnels.
- Conducted SEO research and optimization campaigns including technical audits, keyword strategy, content optimization, and backlink acquisition.
- Developed custom WordPress/WooCommerce plugins including event booking systems with date management, inventory tracking, and seat allocation logic.
- Administered client LAMP stack environments using SSH for server management, debugging, and database query optimization.

Business Development Consultant | March 2024 – August 2024

PENINSULA | Auckland CBD

- Enhanced verbal and written communication skills, improving client interactions and team collaboration.
- Translated complex technical concepts for non-technical clients, enhancing stakeholder communication.
- Operated in a metrics-driven environment, linking technical work to measurable business outcomes (KPIs).

Senior Repair Tech & Manager | July 2016 – Nov 2023

SMASHED IT / ILAB | Auckland

- Managed and trained technical staff in diagnostics, repairs, and operational duties.
- Performed website management, SEO optimization, and ad content creation to drive business growth.
- Authored process guides, training documents, and standard operating procedures (SOPs) for quality control.

Core Competencies & Skills

AI & Machine Learning: LangChain, LangGraph (Multi-Agent Systems), OpenAI API (GPT-4, GPT-4o, text-embedding-ada-002), Anthropic Claude API, Google Gemini API, DeepSeek, Perplexity, RAG (Retrieval-Augmented Generation), FAISS, pgvector, PDF Processing, Web Scraping, LLM Prompt Engineering, Chatbot Development, Semantic Search

Data Engineering: pandas, ETL Pipelines, pdfplumber, BeautifulSoup, Data Cleaning & Enrichment, Regex Pattern Engineering, Batch Processing

Frontend: React 19, TypeScript, JavaScript, Next.js 16, React Native, HTML5, CSS3, Tailwind CSS, Mantine UI, TanStack Query, Zustand, TipTap, Figma

Backend: Node.js, Express.js, Python, PHP, WordPress, WooCommerce Plugin Development, REST APIs, WebSockets, Prisma ORM, Sequelize ORM, Zod Validation, Session Management, Authentication & Authorization

Databases: PostgreSQL (pgvector), MySQL, Redis, DynamoDB

DevOps & Cloud: Docker, Docker Compose, CI/CD, GitHub Actions, AWS (EC2, S3, RDS, Lambda), Google Cloud Platform, Linux Server Administration, SSH, Apache, NGINX, Proxmox VE

Automation & Integtic: n8n.io, BullMQ (Queue Processing), TinyBird (Analytics), Resend API, Nodemailer (SMTP), WooCommerce, Google Merchant Center, Google Search Console, Google Analytics 4, Google Business Profile, Stripe, SEO Optimization

Education

Bachelor of Science, Computer Science | Graduated Feb 2024

University of Auckland | Auckland, New Zealand